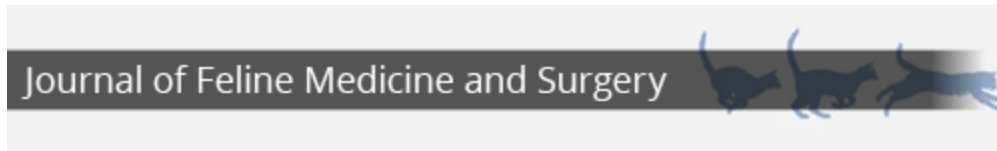


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Feline CKD

Pathophysiology and risk factors — what do we know?

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Abstract

Practical relevance:

Chronic kidney disease (CKD) is one of the most frequently encountered disorders in cats, having increased in prevalence in recent decades. Although the underlying cause is rarely identified, the common final outcome of feline CKD is tubulointerstitial fibrosis. Knowledge of CKD pathophysiology is necessary for optimal individualised patient management, especially with regard to diagnosis and treatment of extrarenal complications.

Patient group:

CKD is most common in senior and geriatric cats, but should be considered in any feline patient with ureterolithiasis, hyperthyroidism, retrovirus infection, systemic hypertension, cardiovascular disease or urinary tract infection.

Evidence base:

Most of our knowledge of the pathogenesis of CKD is extrapolated from human nephrology and experimental animal studies. There is, therefore, a need for further studies in cats. The prevalence of clinical signs in feline CKD is well documented. Several concurrent diseases associated with CKD have also been reported in cats, especially in the geriatric population, but there is no or only limited published evidence demonstrating a cause-and-effect relationship between most of these conditions and CKD. Studies performed over the past 15 years have nevertheless allowed identification of major risk factors (proteinuria, plasma phosphate and plasma creatinine) influencing the progression of feline CKD.

Clinical challenges:

Clinical signs occur in the late stages of renal disease, so populations at higher risk of CKD should be screened routinely. CKD-associated complications (systemic hypertension, secondary renal hyperparathyroidism, hypokalaemia, anaemia, metabolic acidosis) must not be overlooked as they may affect the progression of disease. Disease progression is itself unpredictable and renal function may remain stable for extended periods. Most cats with early CKD do not progress to end-stage CKD before they die.

Audience:

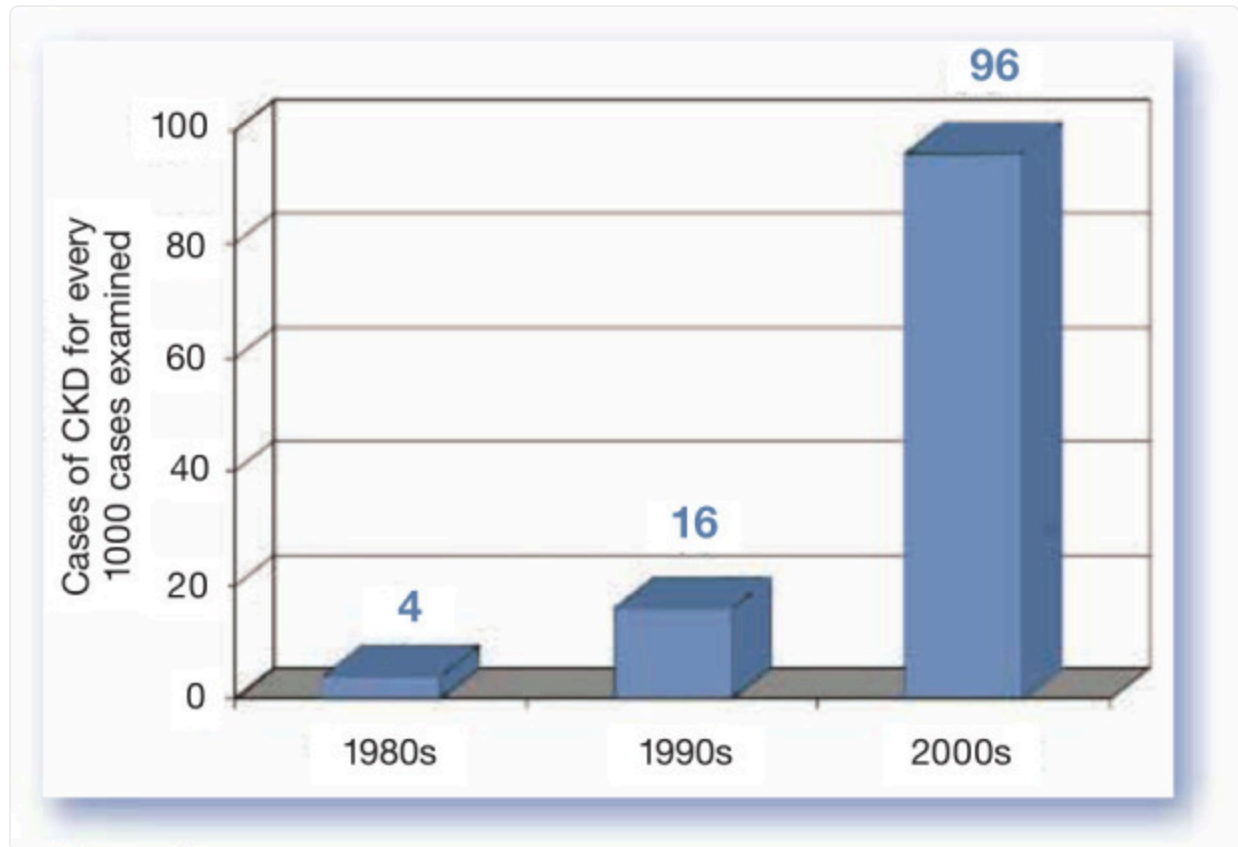
General practitioners play a major role in screening feline patients at risk of development or progression of CKD.

Introduction

Chronic kidney disease (CKD) is the most common renal disease in cats and is defined as structural and/or functional impairment of one or both kidneys that has been present for more than 3 months.¹ CKD may result from any condition that causes progressive and irreversible damage to the kidneys and indeed is intimately associated with an entire spectrum of diseases that occur following initiation of kidney damage. The severity of CKD is denoted by the International Renal Interest Society (IRIS) staging system according to the degree of azotaemia, with stage 1 representing non-azotaemic disease, through to stage 4 representing severe end-stage renal azotaemia.

The prevalence of feline CKD is high and seems to be continuing to rise rapidly ([Figure 1](#)).

Figure 1.



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Prevalence of CKD in recent decades among cats of all ages recorded within the Purdue Veterinary Medical Database^{2,3}

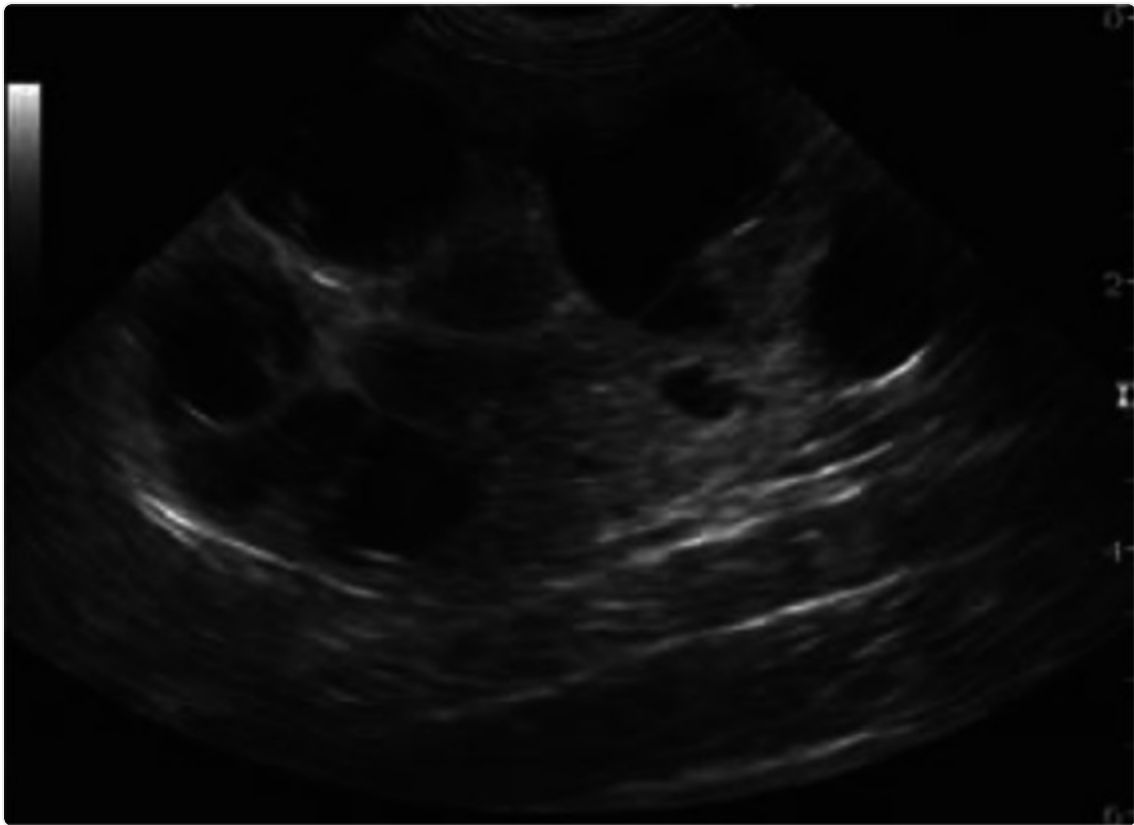
Aetiology of feline CKD

CKD is a non-specific term that does not indicate the cause of kidney damage and/or impaired kidney function. The causes are heterogeneous and most often not identified. They can be divided into two groups – congenital and acquired.

Congenital diseases

Congenital diseases are present at birth and include inherited disorders. The most common genetic disease in cats is autosomal dominant polycystic kidney disease ([Figure 2](#)), which in the past has affected up to 38% of Persian and Persian-cross cats (recent data suggest the prevalence is lower as a result of selective breeding efforts).⁴ Other congenital feline kidney diseases are renal amyloidosis in Abyssinian, Siamese and Oriental cats,⁴ glomerular disease in Abyssinian cats⁵ and juvenile renal dysplasia.⁶

Figure 2.



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Ultrasonographic appearance of a polycystic kidney in a cat

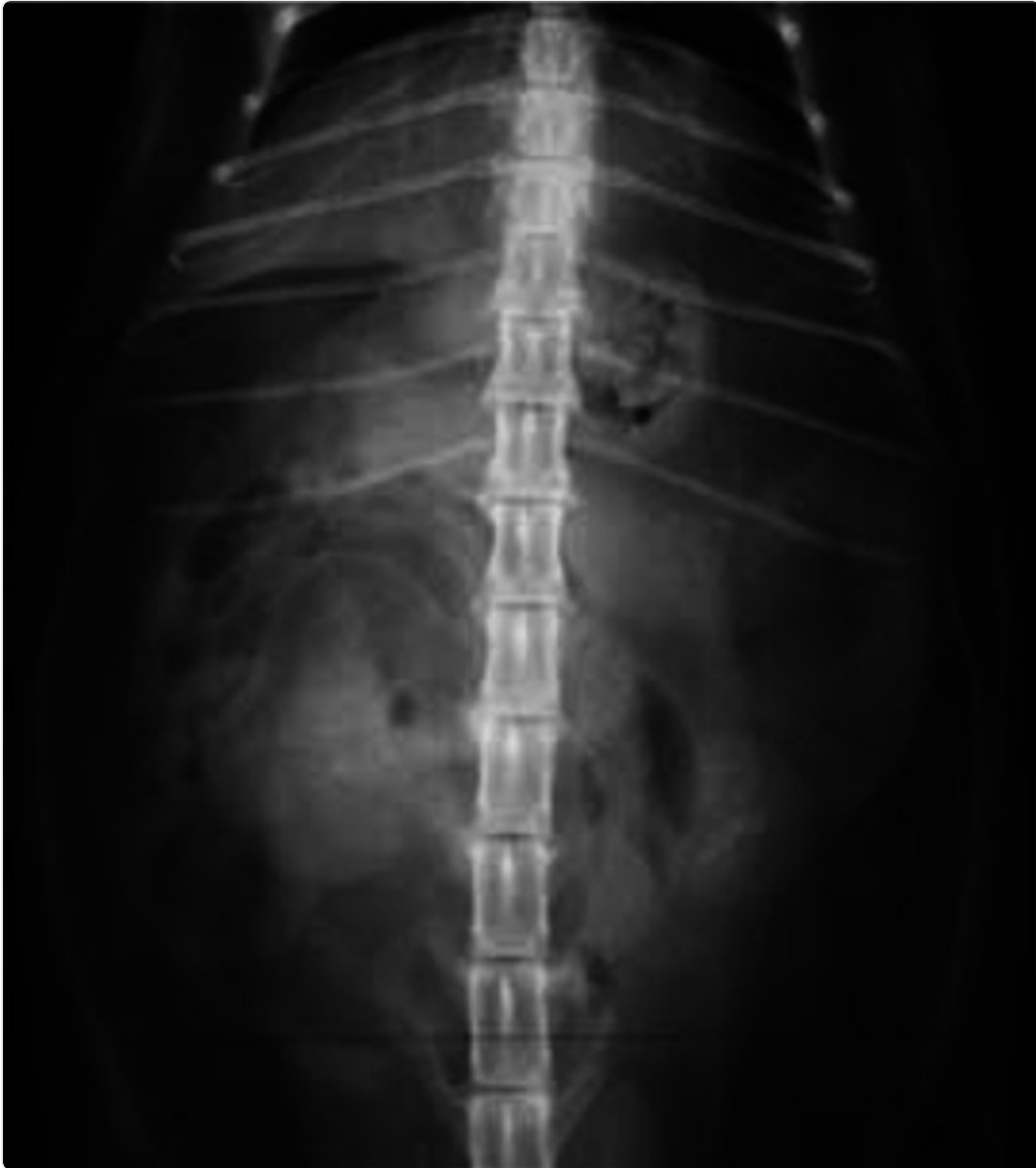
Acquired diseases

Acquired diseases have been identified, or are suspected, in many cases of CKD in cats.

- **Upper urinary tract uroliths** were diagnosed in two studies in 15%⁷ and 29%² of cats with CKD. Azotaemia and hyperphosphataemia are frequent findings in cats with ureteral calculi. Interestingly, 76% of cats with unilateral ureteral calculi are azotaemic and 56% show a small contralateral kidney ([Figure 3](#)),⁸ suggesting that pre-existing renal disease is common in cats with ureterolithiasis. After treatment, serum creatinine remains elevated in approximately half of the cats.⁹ It is, however, difficult to disentangle whether ureterolithiasis is the cause or the consequence of CKD ([Figure 4](#)).
- **Renal lymphoma** ([Figure 5](#)), the most common primary renal neoplasm, may cause CKD. It has been reported in 16% of cats with CKD,^{3,10} and 60% of cats with renal lymphoma are azotaemic.¹¹
- **Hyperthyroidism** is the most common endocrine disease of aged cats and may be observed concurrently with CKD. Based on several studies, the reported prevalence of pre-existing CKD in hyperthyroid cats ranges from 14–40%.¹² The ratio of urinary retinol binding protein (RBP) to creatinine concentrations is increased in cats with hyperthyroidism.¹³ As RBP is freely filtered and normally totally reabsorbed, this increase reflects tubular lesions and dysfunction, and may therefore indicate that hyperthyroidism could induce kidney damage. Weight loss, vomiting, polyuria, polydipsia, proteinuria and low urine specific gravity are frequently encountered in both diseases.
Hyperthyroidism can mask and even exacerbate co-existing CKD, with renal disease becoming apparent in 15.3–39% of cats after treatment of hyperthyroidism.^{12,14,15} Proteinuria is an indicator of all-cause mortality in hyperthyroid cats after treatment but not a mediator of progression of CKD in these patients.¹⁵
- **Infections** might well play a causal role in some cases of CKD. There is some evidence of a link between the development of CKD and feline immunodeficiency virus (FIV)¹⁶ and feline leukaemia virus (FeLV) infections.¹⁷ Experimental FIV infection induces renal lesions¹⁶ and clinical CKD has been reported in 9.3% of FIV-infected cats.¹⁸ Mild to severe proteinuria, but not azotaemia, is also associated with FIV infection and has been reported in 25% of cases.^{16,19} More recently, a previously undescribed paramyxovirus, feline morbillivirus, was also shown to be associated with tubulointerstitial nephritis.²⁰
- **Vaccines** containing viruses grown on Crandell Rees feline kidney cells may induce transient renal inflammation.²¹ They contain some feline kidney cell antigens and stimulate production of autoantibodies in vaccinated cats.²² No evidence, however, exists to demonstrate that these vaccine-induced responses may initiate CKD in cats.
- **Nephrotoxic drugs** (eg, aminoglycosides, non-steroidal anti-inflammatory drugs, antineoplastic agents) or **poisoning** (eg, lily toxicity) can cause acute kidney injury (AKI). Incomplete renal recovery after AKI could lead to CKD and accelerate the progression to end-stage renal disease.

- Exclusive feeding of an inappropriate **diet** (high in protein and acid, depleted in potassium) for several years was shown to induce CKD in healthy cats. [23](#) Ad libitum feeding and increased ash intake could be associated with an increased risk of CKD. [24](#) It has been hypothesised that high salt diets might adversely affect renal function, principally by inducing systemic hypertension. However, a recent study of aged cats fed a high sodium diet for 2 years showed no effect on renal function. [25](#)

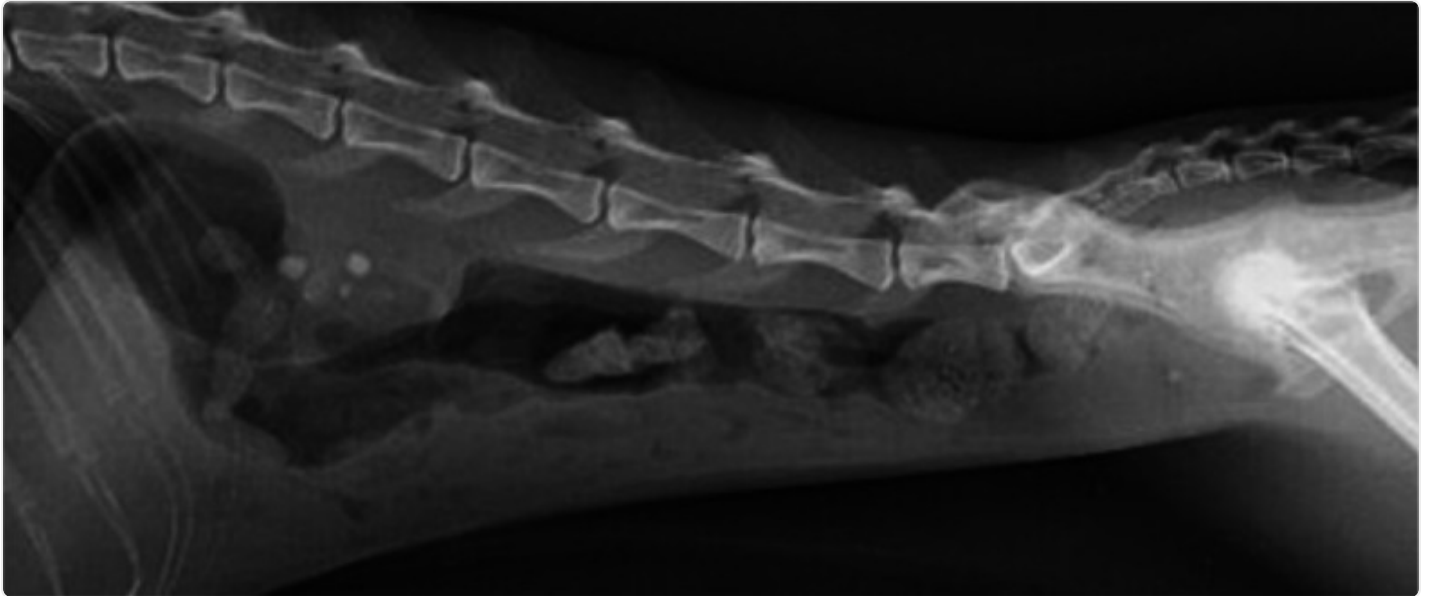
Figure 3.



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'Big kidney-little kidney' syndrome in an azotaemic cat. Abdominal radiographic findings such as these are indicative of ureteral obstruction

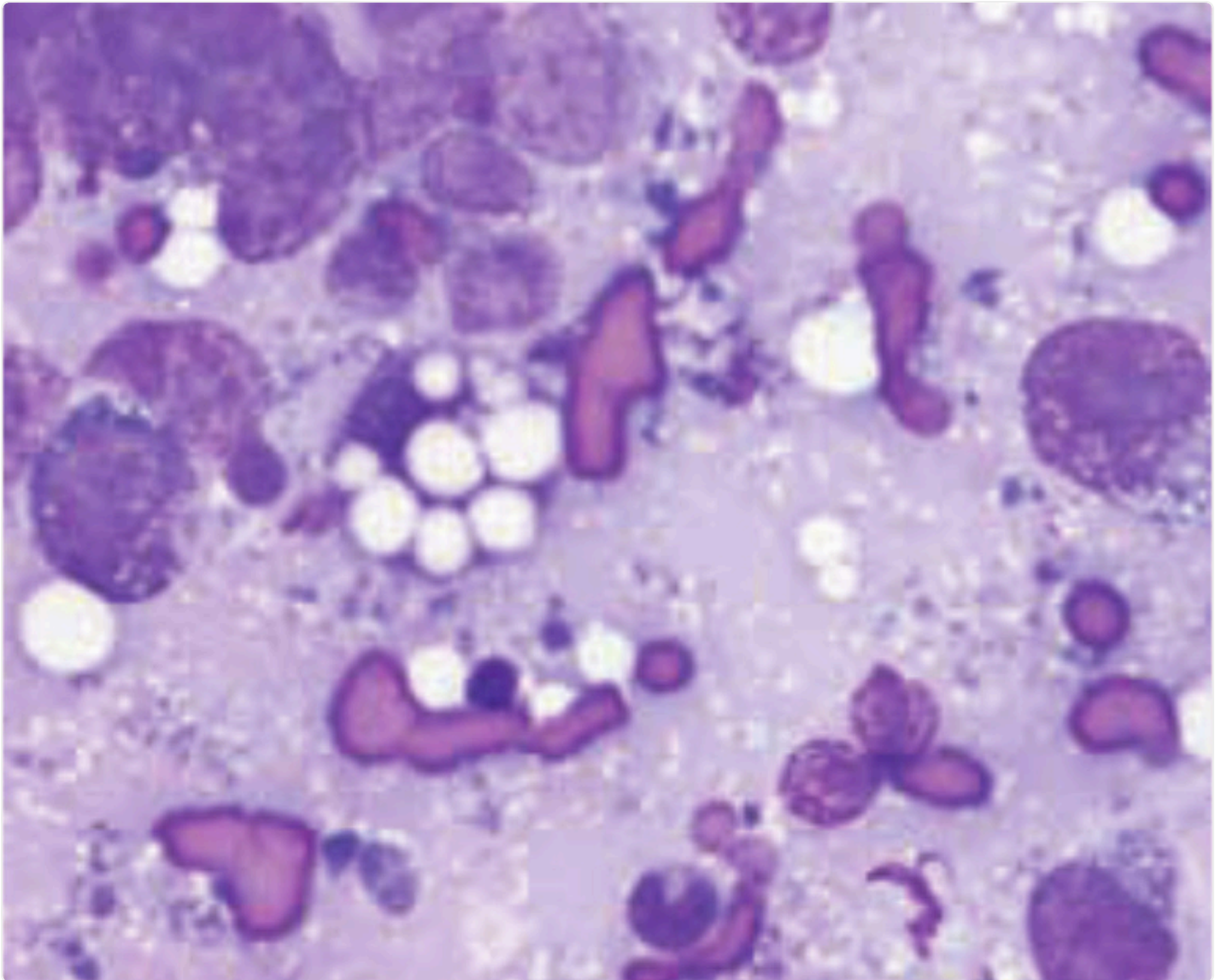
Figure 4.



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Abdominal radiograph showing multiple nephroliths in a cat with advanced CKD

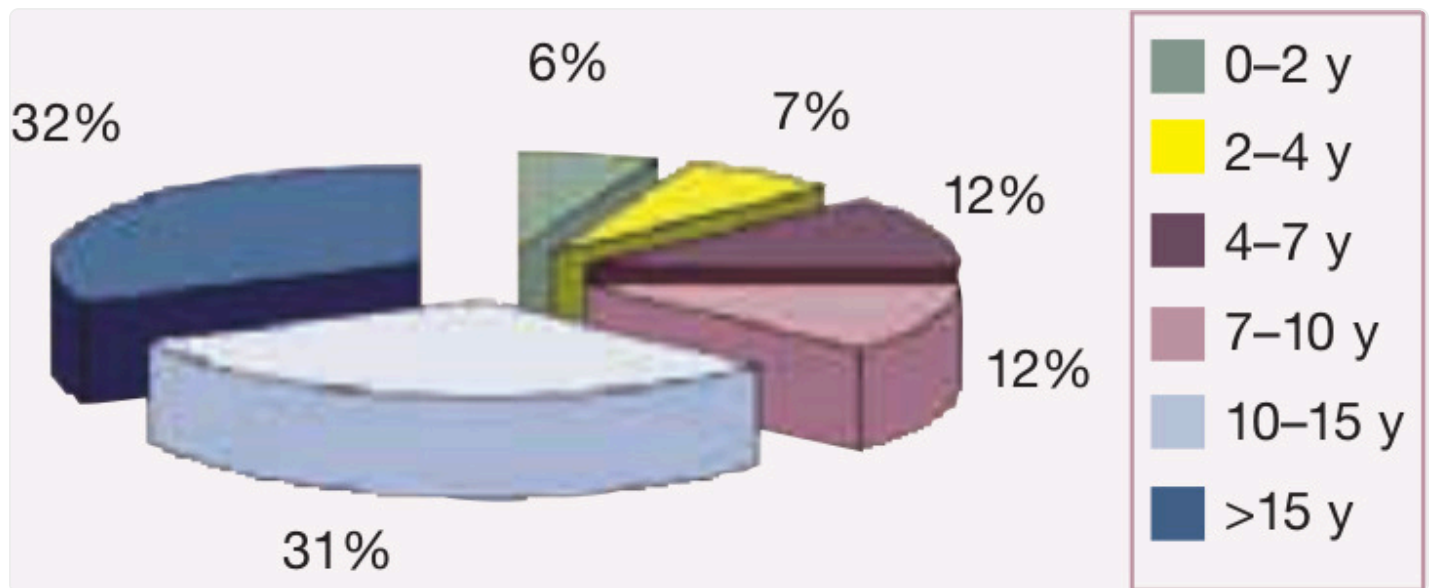
Figure 5.



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Cytology of a fine-needle aspirate from the kidney of a cat with renal lymphoma. Large blastic lymphoid cells typically have a high nucleocytoplasmic ratio, microvacuolated basophilic cytoplasm and a round nucleus with reticulated chromatin and prominent basophilic nucleoli

Figure 6.



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Distribution of 'renal failure' in cats of different ages based on 2228 case records from patients diagnosed at 23 US veterinary colleges³

Ageing

Ageing is considered a risk factor for the development of CKD in cats, with CKD indeed being more frequently diagnosed in senior and geriatric cats (Figure 6).^{3,10,26–28} The mortality rate due to CKD is also greater in cats over 9 years old, and cats of any age that die from non-renal causes frequently have renal lesions identified.²⁶ The lesions are mainly tubulointerstitial,^{28,29} and their severity increases with age.²⁹ In one study, 30.5% of aged, apparently healthy cats became azotaemic within 12 months; only plasma creatinine concentration and proteinuria at presentation were predictive of cats at risk of developing azotaemia within this time period.³⁰ However, cats that developed azotaemia were not older than those that remained non-azotaemic, suggesting that ageing per se is not the primary cause of development of azotaemia. The role of systemic hypertension and proteinuria in the development of azotaemia in ageing cats remains unclear. Systolic blood pressure (SBP) has been reported to be minimally affected by ageing in cats.²⁶ The urine protein:creatinine ratio (UPC) is associated with the development of azotaemia,³⁰ but ageing in cats with CKD is not a risk factor for increasing proteinuria.³¹ One study found that, among cats that died from non-renal causes, those that had tubulointerstitial lesions had a longer life span than those with no renal changes.²⁸ Consequently, it was suggested that ageing-associated renal changes might even be protective.²⁸ This finding requires further investigation, however, as the study in question had several potential limitations.³²

Breed

Breed appears to be a potential risk factor for the development of CKD, with Siamese, Abyssinian, Persian, Maine Coon and Burmese cats reportedly more likely to be affected.^{3,33} The strength of evidence for a breed-associated risk is low, however.

Gender

Gender is not a risk factor for the development of CKD.^{3,10,33,34} However, male cats with CKD are younger and overrepresented between the ages of 9 and 11 years.³⁴

Systemic hypertension

Systemic hypertension (as well as cardiovascular disease and urinary tract infections [UTIs] – see below) is a frequent concurrent disease in aged cats with CKD. These various conditions could also contribute to the development of renal disease, although conclusive

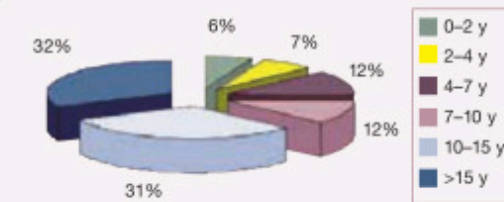


Figure 6 Distribution of 'renal failure' in cats of different ages based on 2228 case records from patients diagnosed at 23 US veterinary colleges³

cause-and-effect relationships have yet to be demonstrated. Systemic hypertension is diagnosed in about 20–65% of cats with CKD.³⁵ Azotaemia is also observed in 61% of hypertensive cats.³⁶ In feline CKD, systemic hypertension is associated with proteinuria but is not predictive of survival.^{31,37} Moreover, it is not associated with interstitial fibrosis or inflammation, suggesting that systemic hypertension is not a primary factor in the progression of

CKD, at least in cats receiving amlodipine therapy.³⁸ However, if cats remain untreated, hypertension might lead to more severe lesions.³⁸

Cardiovascular disease

Cardiovascular diseases resulting from, or independent of, systemic hypertension have been reported in cats with CKD.^{39,40} The prevalence of left ventricular hypertrophy has been estimated to be 46.6% in cats with CKD.³⁹ About 59% of cats with hypertrophic cardiomyopathy (HCM) are azotaemic and 12.7% have CKD. Systolic arterial pressure is also higher in azotaemic than in non-azotaemic cats with HCM.⁴¹ Investigations are needed to document the pathophysiology of concomitant CKD and cardiac disease.

Primary hyperaldosteronism

Primary hyperaldosteronism might also mediate the development of CKD. Mild azotaemia, hypokalaemia and systemic hypertension with associated ocular signs are very common in cats with primary hyperaldosteronism.^{42,43} Moreover, a gradual increase in azotaemia and the development of CKD has been observed in such patients and CKD was reported as a cause of euthanasia.^{42,43} Histopathological renal changes were similar to those observed in humans with hyperaldosteronism.⁴²

Urinary tract infections

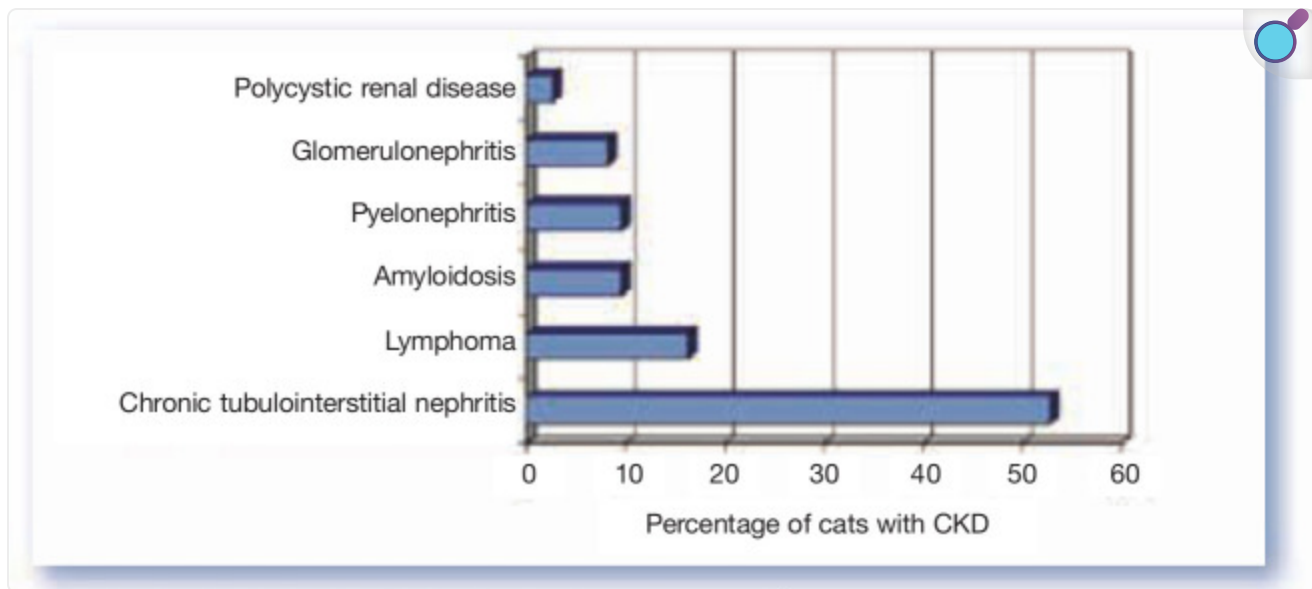
UTIs are considered a risk factor for the development of CKD. Experimental UTI induces chronic pyelonephritis, nephritis and renal fibrosis in cats.⁴⁴ Conversely, in cats with CKD, decreased local and systemic defence mechanisms, urodynamic alterations and changes in the composition of urine might lead to UTI. In feline CKD, the prevalence of UTI is reported to be 17–33%^{3,7,10,45} and that of chronic pyelonephritis has been estimated at 9.5–42%.^{10,46} It has also been reported that two-thirds of cats with UTIs older than 10 years were affected by CKD.⁴⁷ However, in cats with CKD, no association is observed between UTI and markedly elevated serum creatinine concentration or other urine variables.^{7,45}

Pathogenesis of feline CKD

Tubulointerstitial fibrosis appears to be the most common final outcome of feline CKD (Figures 7 and 8).^{2,3,10,48} Primary glomerular diseases are rare – glomerular disease develops secondarily to systemic disease in most cases, specifically neoplastic (eg, leukaemia, lymphoma), infectious and non-infectious inflammatory disorders (eg, pancreatitis, immune-mediated diseases).⁴⁹ The presence of interstitial fibrosis is the strongest histomorphometric predictor of plasma creatinine concentration.³⁸ Tubular damage occurs early in the course of disease, before elevations in serum

urea and creatinine concentrations.⁵⁰ Although cats are able to maintain urine concentrating ability after subtotal nephrectomy,⁵¹ urine specific gravity decreases as plasma creatinine increases in cats with CKD,^{52,53} suggesting a close relationship between glomerular and tubular function during the progression of feline CKD.

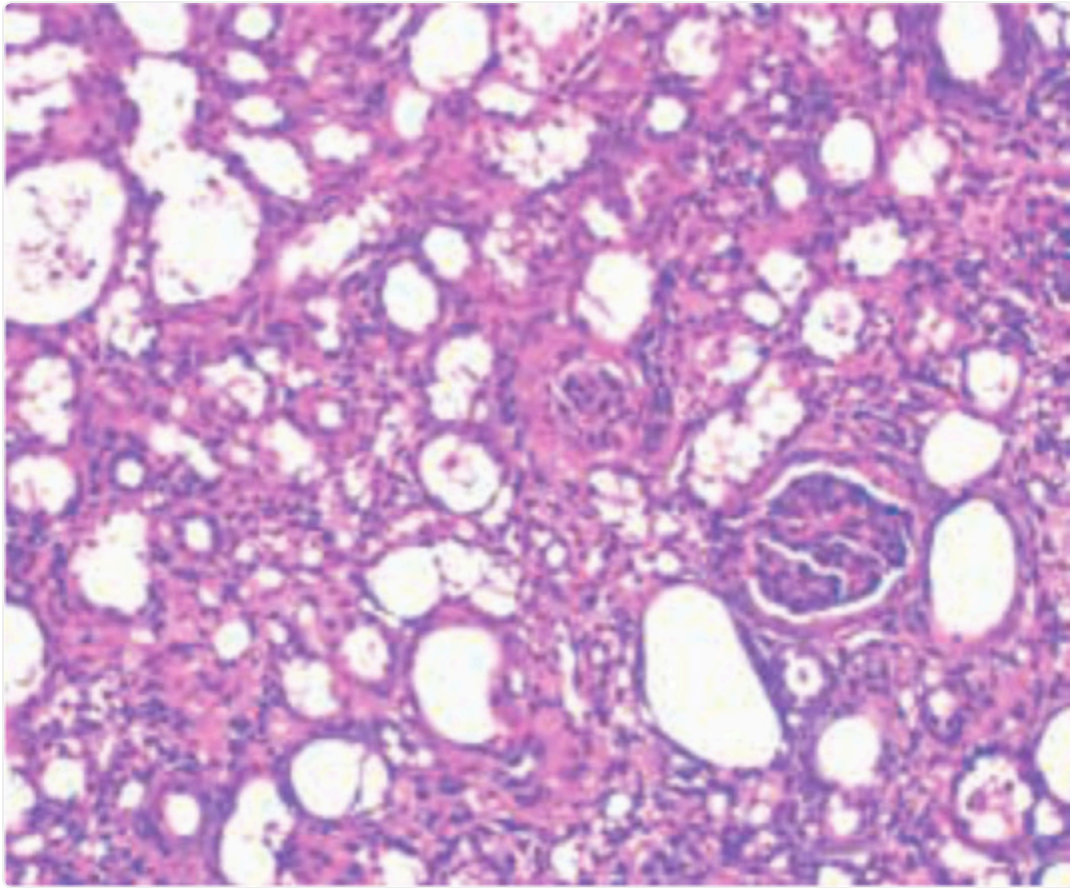
Figure 7.



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Distribution of renal lesions in 64 cats with CKD¹⁰

Figure 8.



[Open in a new tab](#)

Renal histology in a cat with end-stage CKD. Note the interstitial inflammation, fibrosis and tubule dilation. H&E, original magnification x200. Courtesy of Dr A Poujade and Dr C Dally, Laboratoire d'Anatomie Pathologique Vétérinaire du Sud-Ouest, Toulouse, France

Although most of the factors that initiate kidney damage remain unknown in cats, it is hypothesised that the process is similar to that described in other species (see box below).⁵⁴

Sustained and chronic kidney injury induces irreversible inflammation with infiltration of inflammatory cells, which in turn serves as a primer that initiates renal fibrogenesis. Inflammatory cells produce profibrotic cytokines (eg, transforming growth factor [TGF- β 1]) which promote fibrogenesis by activation of matrix-producing cells. Urine excretion of TGF- β 1 is increased in feline CKD.⁵⁵

Stages of tubulointerstitial fibrosis

Sustained and chronic kidney injury

Infiltration of inflammatory cells

Fibroblast activation and expansion from various sources

Production and deposition of extracellular matrix

Fibrosis

Duration of injury



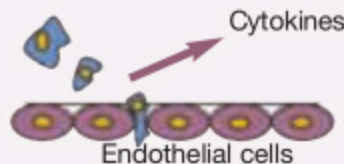
Reversibility of fibrosis



Injury



Tubular epithelial cells



Endothelial cells

Interstitial fibroblasts

Myofibroblasts

Pericytes, fibrocytes, epithelial-mesenchymal transformation

Collagen
Fibronectin

Activated fibroblasts, also referred to as myofibroblasts, produce large amounts of extracellular matrix components (collagen, fibronectin). Other sources of myofibroblasts are pericytes, fibrocytes and tubular epithelial cells that undergo epithelial-mesenchymal transformation. Myofibroblasts express α -smooth muscle actin and vimentin. In cats with CKD, interstitial expression of these markers correlates with the extent of tubulointerstitial lesions, interstitial fibrosis, interstitial fibronectin deposition and renal dysfunction.^{56,57} Chronic hypoxia is often also present and is associated with increased oxidative stress,⁵⁴ which is likely to be important in the pathogenesis of feline CKD.^{58,59}

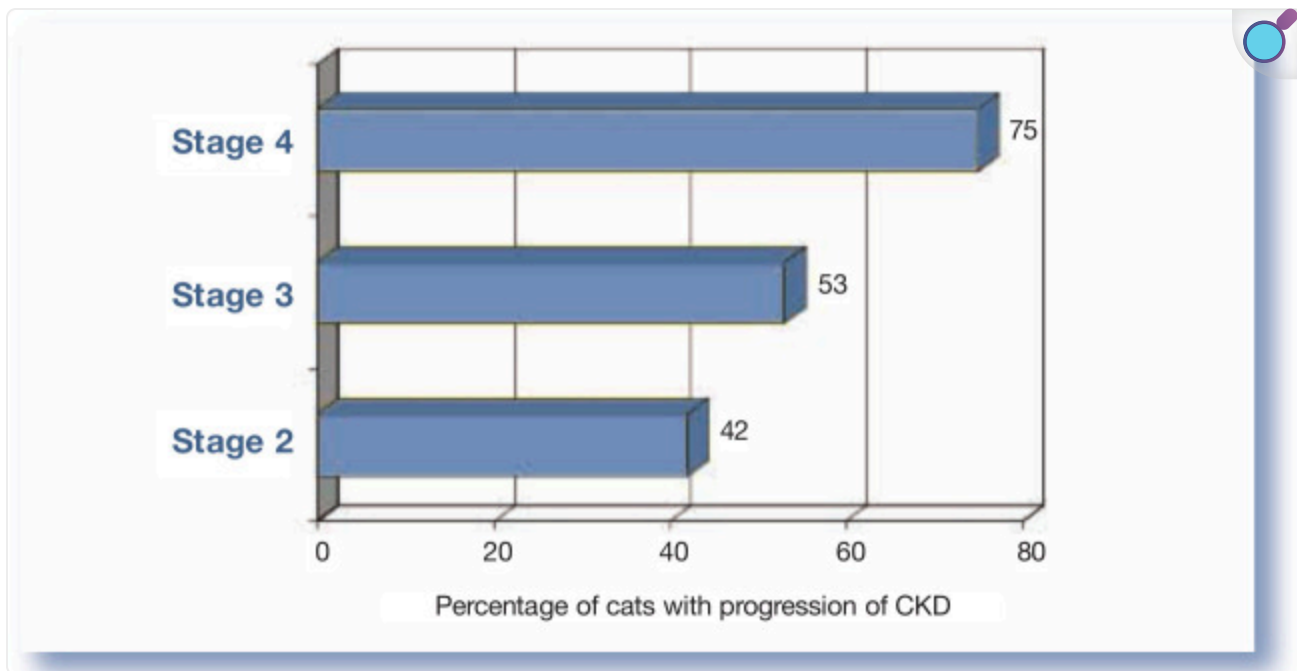
The renin–angiotensin–aldosterone system (RAAS) is additionally activated in cats with naturally occurring or experimental CKD.^{39,60–63} Angiotensin II induces vasoconstriction of the efferent arteriole (and consequently glomerular hypertension) and accumulation of extracellular matrix mediated by TGF- β 1.⁶⁴ In cats with CKD, angiotensin-converting enzyme (ACE) inhibitors decrease glomerular capillary pressure, the ratio of efferent to afferent arteriolar vascular resistance and proteinuria, but have not been shown to improve histopathological changes in the kidneys or survival times.^{46,65} Insufficient statistical power might, however, explain the lack of effect on survival. Aldosterone could also contribute to the development of CKD via several pathogenic mechanisms (see box on the right).⁶⁶

Progression of feline CKD

The progression of CKD is postulated to result from a self-perpetuating vicious cycle. It is attributed to maladaptive functional and structural changes in surviving (or remnant) nephrons after renal injury; in cats with experimental CKD, these are principally increases in single nephron glomerular filtration rate (GFR), glomerular pressure and glomerular volume.⁶⁷

Because of the large reserve capacity of the kidneys, at least 75% of renal function needs to be impaired before azotaemia is detected.⁶⁸ In cats, GFR remains stable for up to 1 year after induction of experimental CKD.^{69–71} Cats with spontaneous CKD show an unpredictable time course of the disease, with extended periods of stable renal function before decompensation (Figures 9 and 10).^{27,40,72,73} In one study, for example, 81% and 37% of cats with stages 2 and 3 CKD, respectively, did not progress to stage 4 before they died.⁷³ The median survival time for advanced stage 2 (serum creatinine 205–250 μ mol/l, 2.3–2.8 mg/dl), 3 (serum creatinine 251–440 μ mol/l, 2.9–5.0 mg/dl) and 4 (serum creatinine >440 μ mol/l, >5.0 mg/dl) was estimated at 1151, 778 and 103 days, respectively, but large individual variation was observed.³³

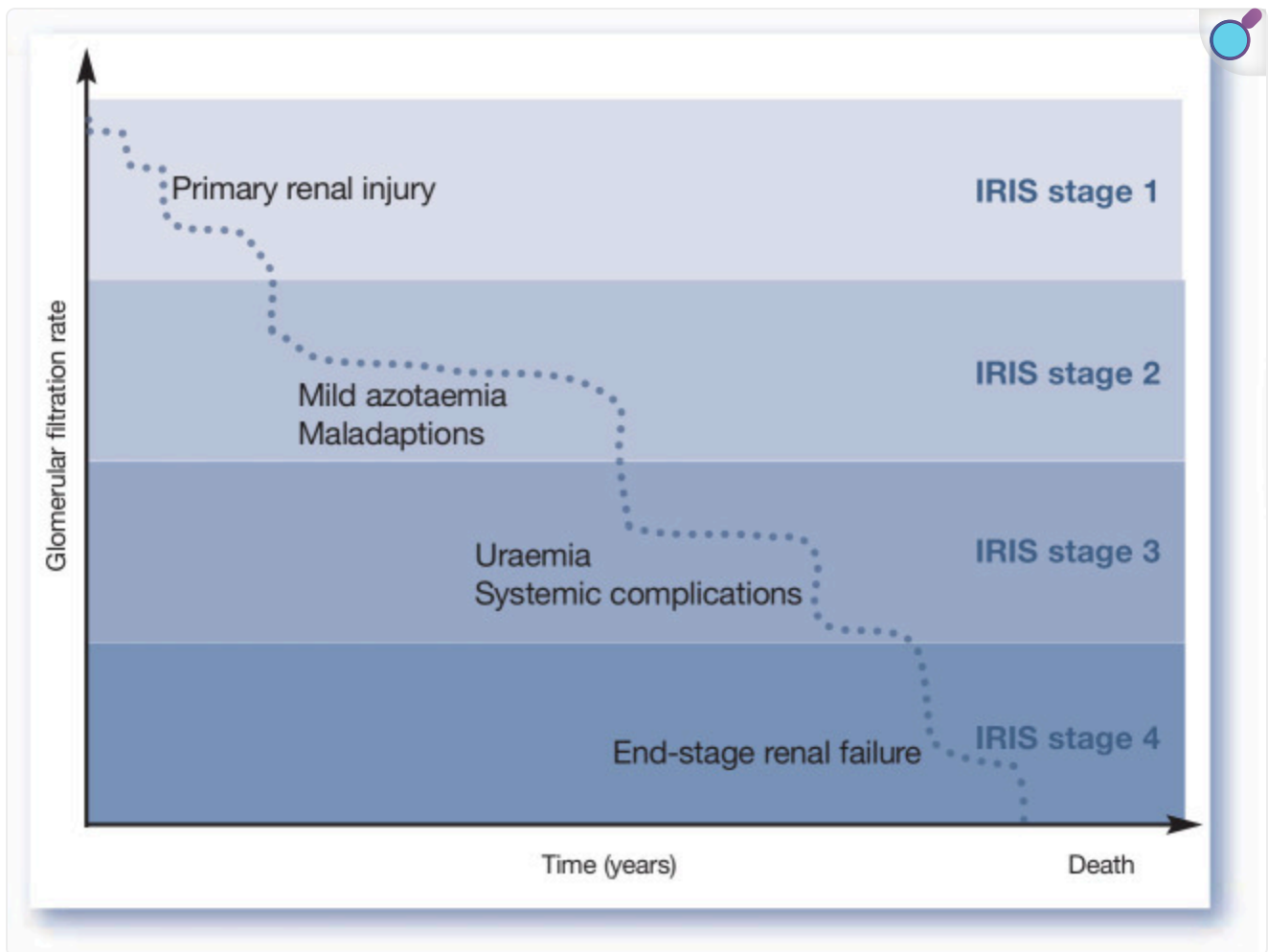
Figure 9.



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Percentage of cats with progression of CKD (ie, $\geq 25\%$ increase in plasma creatinine concentration relative to the concentration at diagnosis) within 1 year of diagnosis, according to IRIS stage of renal disease ⁷³

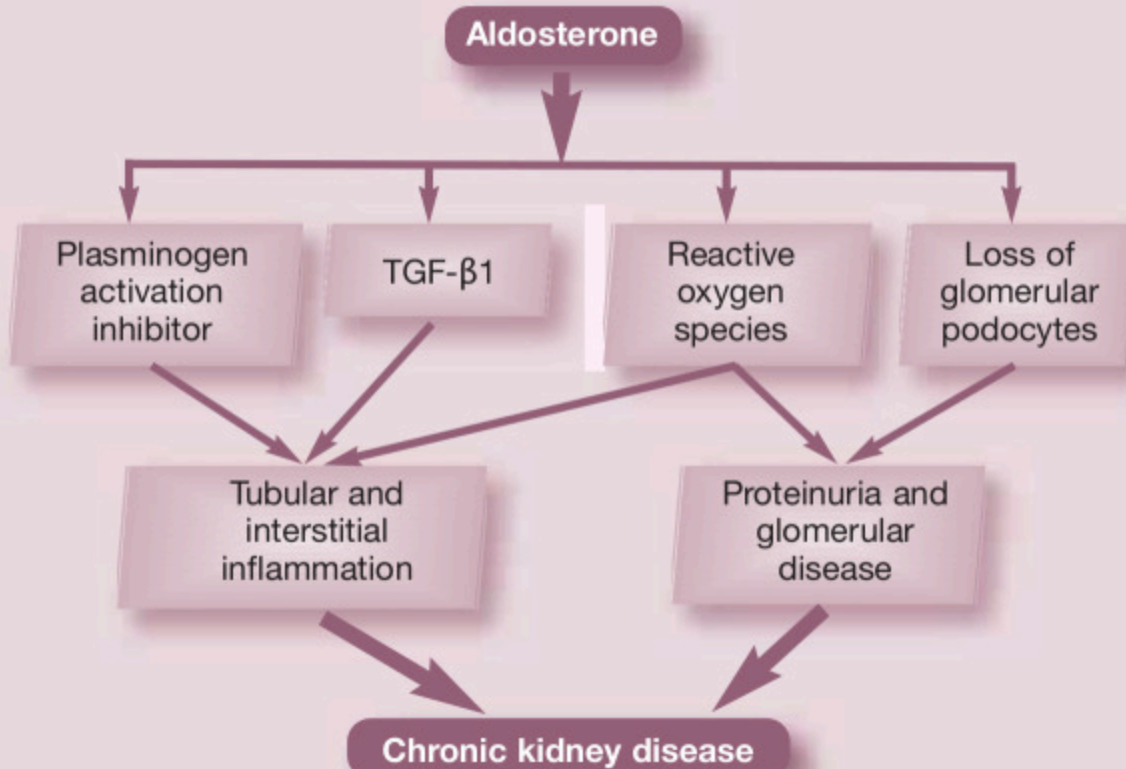
Figure 10.



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Decline in glomerular filtration rate (dotted line) during the progression of CKD from IRIS stage 1 to stage 4. The rate of progression varies according to the nature of the kidney damage and among individual cats. Cats in stages 2 and 3 can have stable renal function for sustained periods until decompensation

Adverse renal effects of aldosterone in CKD



Risk factors for progression

Factors affecting patient survival time and disease progression in cats with CKD have been evaluated in different studies (see accompanying boxes).^{26,31,33,53,73,74} Increased plasma concentrations of creatinine,^{31,53,73} phosphate^{33,53,73} and urea,^{53,73} increased UPC,^{31,53,73} decreased blood haemoglobin concentration and haematocrit,^{31,53,73} and increased blood leukocyte counts⁵³ are associated with a shorter survival time in cats with CKD. For example, an increase in UPC of 0.1 in cats with stage 2 CKD was associated with a 23% increase in the risk of progression within 1 year of diagnosis.⁷³ In one study, laboratory data obtained from cats with CKD at initial evaluation and later within 1 month before death were retrospectively compared.⁷⁴ An increase in plasma creatinine and UPC and a decrease in packed cell volume (PCV) were observed.

However, no single variable has been identified that reliably predicts the survival time in a given cat with CKD.^{53,73} Survival time in relation to various clinical criteria (Table 1) is also very variable.³³

Factors shown not to affect CKD progression

- ❖ Breed⁵³
- ❖ Plasma calcium³³
- ❖ Plasma bicarbonate³³
- ❖ Plasma potassium^{33,53}
- ❖ Presence of UTI⁷³
- ❖ Presence of systemic hypertension^{31,73}

Factors with conflicting results regarding influence on CKD progression

- ❖ Age^{31,33}
- ❖ Body weight^{53,73}
- ❖ Plasma albumin^{33,73}

Factors found to be associated with decreased survival time in CKD patients

- ❖ High UPC^{31,53,73}
- ❖ High plasma creatinine^{31,53,73}
- ❖ High plasma phosphate^{33,53,73}
- ❖ High plasma urea^{53,73}
- ❖ High blood leukocyte counts⁵³
- ❖ Low blood haemoglobin and haematocrit^{31,53,73}

No single variable has been identified that reliably predicts the survival time in a given cat with CKD.



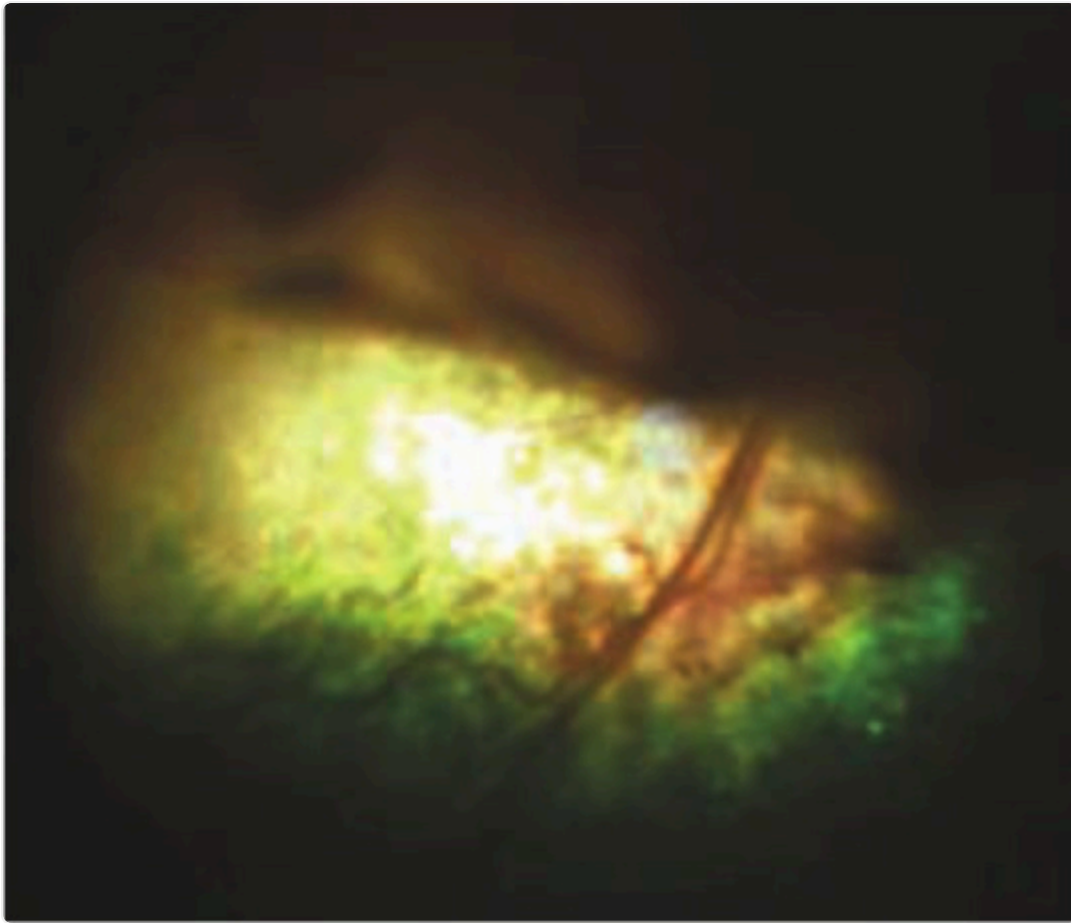
Pathophysiology of secondary changes occurring in feline CKD

The clinical signs and complications of CKD progress with declining GFR due to decreased elimination of, and hence overexposure to, endogenous substances (so-called uraemic toxins). These complications are also considered in themselves to be major contributors to the progression of renal disease, and increase the risk for morbidity and mortality in patients with CKD.

Systemic hypertension

Systemic hypertension is commonly associated with CKD. The pathogenesis of hypertension in cats with CKD is multifactorial and includes impaired excretion of sodium, RAAS activation, stimulation of the sympathetic nervous system, arterial structural changes, endothelial dysfunction and oxidative stress.³⁵ Systemic hypertension may adversely affect renal function by inducing glomerular hypertension and proteinuria; it also increases the risk of other end-organ damage (ie, ocular, neurological and cardiac) ([Figure 11](#)).⁷⁵

Figure 11.



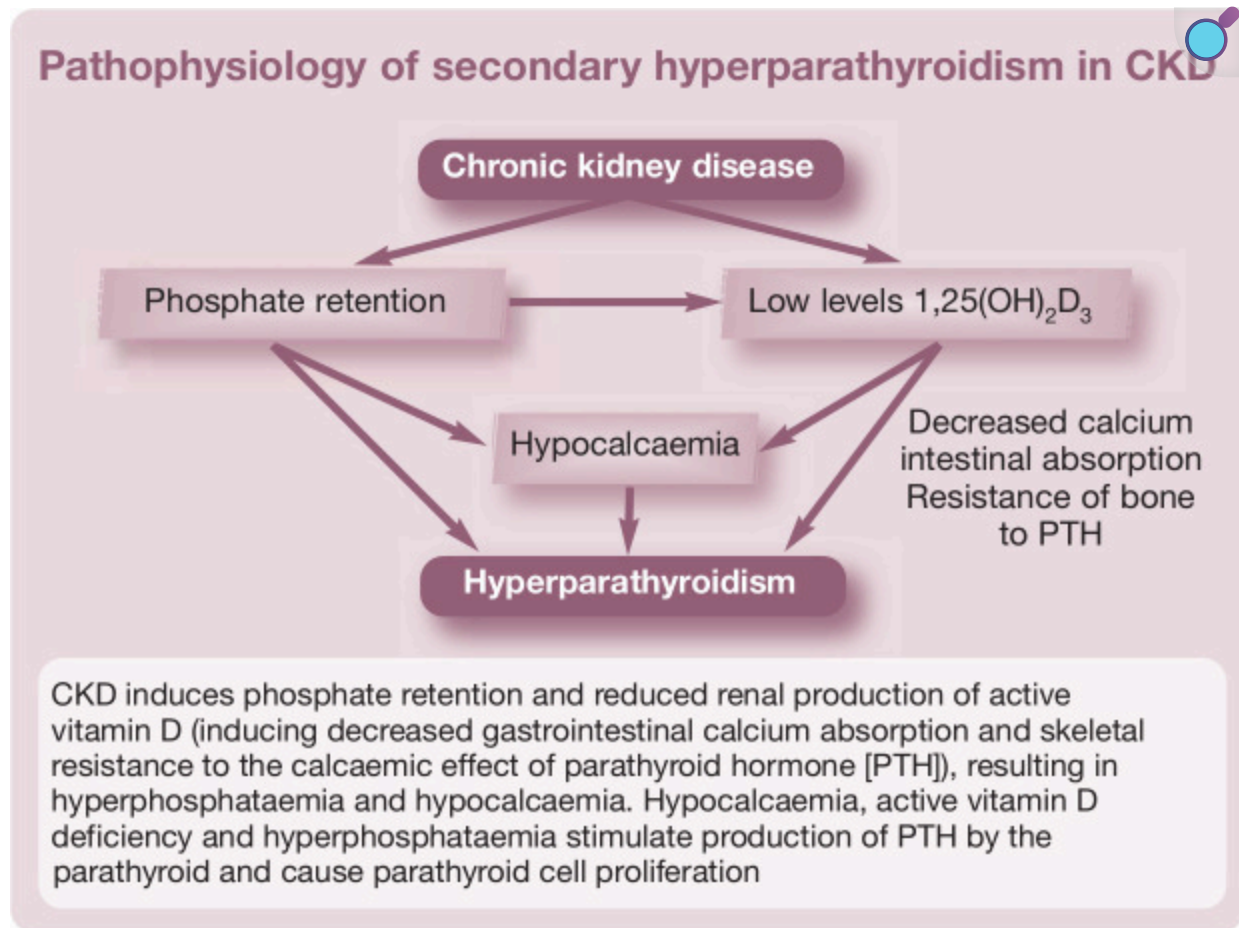
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Dorsal retinal detachment in a cat with systemic arterial hypertension

Renal secondary hyperparathyroidism

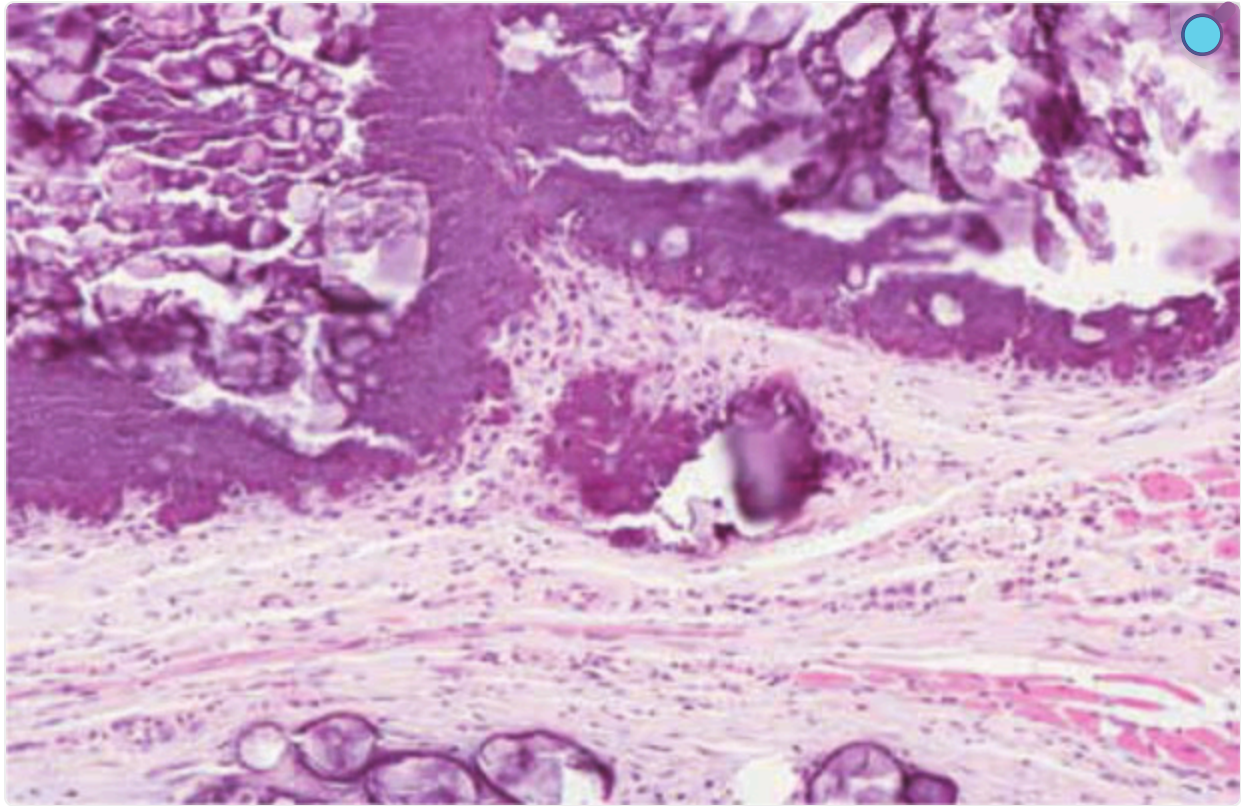
Renal secondary hyperparathyroidism (RHPH) is an adaptive and ultimately maladaptive process that develops in response to declining kidney function, phosphate retention and failure to bioactivate vitamin D (see box above).⁷⁶ It is well recognised that phosphate retention is a major contributor to the progression of CKD. Hyperphosphataemia and increased plasma parathyroid hormone (PTH) concentrations are observed in 13–100% and 47–100%, respectively, of cats with CKD.^{3,53,77} RHPH is particularly common in stages 3 and 4 CKD, as the severity of RHPH increases with the degree of azotaemia.^{3,53,77} Plasma fibroblast growth factor 23 (FGF-23) is a

phosphaturic hormone involved in the pathogenesis of RHPTH in humans. In cats with CKD, plasma FGF-23 is strongly correlated with plasma creatinine concentration and PTH, whereas there is an inverse relationship between FGF-23 and GFR.^{78,79}



Clinical signs induced by RHPTH may result from calcium phosphate precipitation into the tissues (Figure 12) but are relatively uncommon in cats with CKD. Metastatic paw calcification⁸⁰ and osteoporosis¹⁰ have been reported.

Figure 12.



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Soft tissue mineralisation in the tongue of a cat with end-stage CKD. H&E, original magnification x200

Hypokalaemia

Hypokalaemia is a common finding in cats with CKD (prevalence 18–30%), especially those in IRIS stages 2 and 3.^{3,10,26,27,52,81} CKD is also the most common cause of hypokalaemia in cats. Cats with severe hypokalaemia are 3.5 times more likely to have CKD.⁸² Hypokalaemia in feline CKD may result from multiple causes (see box below). It may also develop in renal-impaired cats fed a high protein diet.⁸³ Potassium depletion may induce metabolic acidosis, kidney dysfunction (hypokalaemic nephropathy), muscle weakness and cardiac arrhythmia.^{81,84} Therefore, CKD could lead to potassium depletion, which would in turn contribute to progression of the renal disease and worsening of the clinical condition.

By contrast, hyperkalaemia is uncommon in cats with CKD (prevalence 5.8%),^{46,53} except in end-stage oliguric CKD.⁸¹

Anaemia of renal disease

A normochromic, normocytic, hypoproliferative anaemia is frequently diagnosed in cats with CKD, especially in end stages of the disease;^{85,86} about 30–65% of cats with CKD develop anaemia during progression of their renal disease.⁸⁶ The PCV decreases as the severity of CKD increases.^{52,53} CKD-induced anaemia results mainly from inadequate production of erythropoietin by the peritubular fibroblast type-1 interstitial cells. Other contributing factors to anaemia include malnutrition (eg, iron deficiency), shortened red blood cell life span, myelofibrosis, metabolic alterations affecting erythropoiesis, uraemic toxins, hyperparathyroidism and gastrointestinal blood loss.⁸⁶ Anaemia adversely affects the quality of life of cats with CKD.⁸⁶

Potential causes of hypokalaemia in cats with CKD

Increased loss of potassium

- ✦ Decreased renal potassium reabsorption
- ✦ Diet-induced hypokalaemic nephropathy
- ✦ Renal tubular acidosis
- ✦ Hyperaldosteronism

Translocation of potassium from extracellular to intracellular fluid

- ✦ Hypothermia

Decreased intake of potassium

- ✦ Anorexia

Adapted from DiBartola and Autran De Morais⁸¹

Metabolic acidosis

Metabolic acidosis is also frequent, with a prevalence of 22–88% dependent on the severity of CKD.^{3,10,52} Metabolic acidosis generally occurs late in the time course of CKD.⁷² An increase in anion gap due to low plasma bicarbonate and chloride concentrations is observed,⁵² with development of hypochloraemia preceding acidosis.⁷² Metabolic acidosis may result from potassium depletion,⁸⁴ but hypokalaemic cats are generally able to maintain normal venous blood pH and bicarbonate concentration.⁵² Renal ammoniogenesis, the main renal process for urinary hydrogen ion excretion, could also be impaired in CKD.⁵² Hypochloraemia may result from the loss of chloride through the gastrointestinal tract due to gastric hyperacidity or vomiting.⁷²

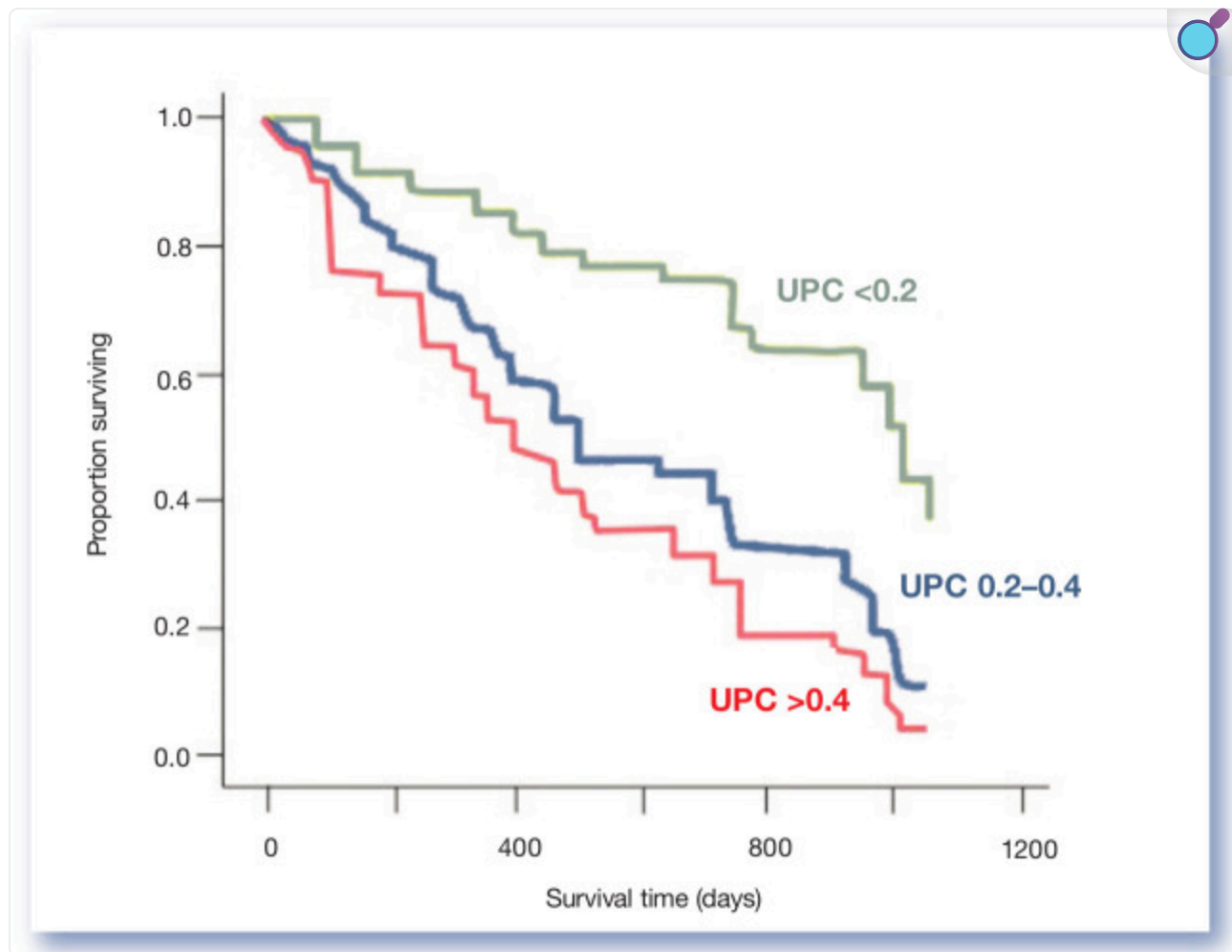
Metabolic acidosis promotes many adverse clinical effects in the renal patient, including anorexia, vomiting, lethargy, weakness and malnutrition.²⁷

Role of proteinuria in disease progression

Proteinuria is a sign of kidney damage, but also a strong indicator for progression of CKD.^{87,88} Small proteins, such as albumin, are filtered by the glomeruli and reabsorbed by the proximal tubule cells by receptor-mediated endocytosis. Dysfunction of these two processes results in proteinuria. Proteinuria accelerates progression of CKD by induction of tubular chemokine expression and complement activation; this, in turn, leads to inflammation and fibrogenesis. Proteinuria is associated with interstitial fibrosis and glomerular hypertrophy,³⁸ and might, therefore, be a marker of tubular dysfunction associated with tubulointerstitial fibrosis.

Proteinuria in cats with naturally occurring CKD is generally mild compared with their human counterparts,^{31,34,46} with 90% and 49% of cats with CKD having a UPC of <1.0 and <0.25, respectively.³¹ The severity of proteinuria, however, has prognostic significance in terms of survival time (Figure 13).^{31,46} Despite proteinuria being generally mild, the ACVIM consensus statement on the treatment of proteinuria recommends therapeutic intervention when UPC $\geq$ 0.4 in cats with CKD causing azotaemia.⁸⁷ Plasma creatinine concentration and SBP are positively correlated with increasing UPC.³¹

Figure 13.



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Effect of proteinuria on survival in cats with CKD. ³¹ The hazard ratio for death or euthanasia was 2.9 and 4.0 for a UPC of 0.2–0.4 and >0.4, respectively, compared with the baseline group (UPC <0.2)

Whether proteinuria is contributing to CKD progression or whether proteinuric CKD is intrinsically more rapidly progressive remains unknown.

Major clinical signs of feline CKD

The prevalence of clinical signs in cats with CKD is presented in [Tables 2](#) and [3](#).

Table 2.

Prevalence of clinical signs detected by owners in cats with CKD

Prevalence [*]	Clinical sign(s)
Very high (>75%)	Lethargy Anorexia Polydipsia Weight loss
High (40–75%)	Depression Vomiting Weakness Polyuria
Moderate (10–39%)	Dysphagia, oral discomfort and buccal cavity lesions Halitosis Constipation
Low (<10%)	Haematuria Dysuria Diarrhoea Neurological signs

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^{*}Maximum percentage of cases showing the clinical sign.

The prevalence can vary between studies^{3,10,26,45,46} and according to the stage of renal disease. Most of the animals included in these studies were cats with advanced-stage CKD

Table 3.

Prevalence of physical examination findings in cats with CKD

Prevalence [*]	Clinical sign(s)
High (40–75%)	Dehydration Cachexia and weight loss
Moderate (10–39%)	Periodontal disease, gingivitis and oral ulceration Hair coat in poor condition Cardiac murmur Large kidneys Small irregular kidneys Palpable goitre Tachycardia Hypothermia Tachypnoea Pallor of the mucous membranes
Low (<10%)	Hyperthermia Retinal lesions

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*Maximum percentage of cases showing the clinical sign.

The prevalence can vary between studies^{[3,10,26,45,46](#)} and according to the stage of renal disease. Most of the animals included in these studies were cats with advanced-stage CKD

Table 1.

Survival time in relation to various clinical criteria

Criteria	Median survival (days)
Diagnosis of CKD	771
Onset of weight loss	401
Start of subcutaneous fluids	142
Onset of anaemia (PCV <25%)	100
>25% weight loss	83
Clinical decompensation	40
Anaemia intervention	25

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PCV = packed cell volume

Adapted from Boyd et al [33](#)

Clinical signs may be non-specific and manifest late in the disease process, once major renal tissue damage has occurred. Many cats remain almost symptom-free until they reach advanced stages of CKD. The frequency of clinical signs is similar in IRIS stages 2 and 3, but stage 4 is associated with increased frequency and severity of signs. [46](#)

Polyuria and polydipsia represent the earliest and most common clinical manifestations of CKD. [27](#) Cats with CKD are more likely to have had polydipsia and polyuria in the year before they are diagnosed. [89](#) Polydipsia, which is a compensatory response to polyuria, is more easily recognised by owners. In advanced stages, polyuria results in dehydration as fluid loss from the body exceeds fluid intake.

Gastrointestinal signs are the most prominent indicators of uraemia and probably represent the main reason for the owner to consult a veterinarian. Hypergastrinaemia, reported in cats with CKD, [90](#) may induce gastric hyperacidity, uraemic gastritis, gastrointestinal bleeding, anorexia and

vomiting. While diarrhoea is uncommon, constipation is more frequent in cats with CKD than in dogs. It is attributable to dehydration but also to the use of phosphate binding agents.⁹¹

Other reported uraemic clinical complications are lethargy, weakness, depression, tremor, myoclonus, seizures, myopathies, uraemic pericarditis and pneumonitis, hypothermia, anaemia and renal osteodystrophy. These disorders are all multifactorial and their pathogenesis has not generally been investigated in cats.

Footnotes

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Key Points

- Whenever CKD is diagnosed in a cat, particular effort should be made to identify underlying conditions that may have a specific treatment.
- Most often, the cause of CKD remains unidentified and only signs of a non-specific tubulointerstitial fibrosis can be detected at the time of diagnosis.
- Age, breed, systemic hypertension, cardiac disease and urinary tract infection are all considered to be risk factors for CKD, although definitive evidence is lacking.
- Whatever its origin, sustained renal injury leads to self-perpetuated maladaptive changes that drive CKD progression.
- Tubulointerstitial fibrosis is the most common histopathological finding in cats with CKD.
- The time course of CKD is typically unpredictable in cats. However, selected clinicopathological variables and clinical criteria may be predictive of a shorter survival time. Notably, the magnitude of proteinuria has negative prognostic significance.
- Cats with CKD should be individually screened for systemic arterial hypertension, renal secondary hyperparathyroidism, hypokalaemia and metabolic acidosis. Appropriate treatments should be implemented on a case-by-case basis.
- Clinical signs of CKD appear late in the course of the disease and are mostly non-specific.

References

1. Bartges JW. Chronic kidney disease in dogs and cats. *Vet Clin North Am Small Anim Pract* 2012; 42: 669–692. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
2. Ross SJ, Osborne CA, Kirk CA, Lowry SR, Koehler LA, Polzin DJ. Clinical evaluation of dietary modification for treatment of spontaneous chronic kidney disease in cats. *J Am Vet Med Assoc* 2006; 229: 949–957. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
3. Lulich JP, Osborne CA, O'Brien RD, Polzin DJ. Feline renal failure: questions, answers, questions. *Compend Contin Educ Pract Vet* 1992; 14: 127–152. [[Google Scholar](#)]
4. Lees GE. Congenital kidney disease. In: Bartges J, Polzin DJ. (eds). *Nephrology and urology of small animals*. Ames: Blackwell Publishing, 2011, pp 568–576. [[Google Scholar](#)]
5. White JD, Norris JM, Bosward KL, Fleay R, Lauer C, Malik R. Persistent haematuria and proteinuria due to glomerular disease in related Abyssinian cats. *J Feline Med Surg* 2008; 10: 219–229. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
6. Aresu L, Zanatta R, Pregel P, Caliarì D, Tursi M, Valenza F, et al. Bilateral juvenile renal dysplasia in a Norwegian Forest cat. *J Feline Med Surg* 2009; 11: 326–329. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
7. Bailiff NL, Westropp JL, Nelson RW, Sykes JE, Owens SD, Kass PH. Evaluation of urine specific gravity and urine sediment as risk factors for urinary tract infections in cats. *Vet Clin Pathol* 2008; 37: 317–322. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
8. Kyles AE, Hardie EM, Wooden BG, Adin CA, Stone EA, Gregory CR, et al. Clinical, clinicopathologic, radiographic, and ultrasonographic abnormalities in cats with ureteral calculi: 163 cases (1984–2002). *J Am Vet Med Assoc* 2005; 226: 932–936. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
9. Kyles AE, Hardie EM, Wooden BG, Adin CA, Stone EA, Gregory CR, et al. Management and outcome of cats with ureteral calculi: 153 cases (1984–2002). *J Am Vet Med Assoc* 2005; 226: 937–944. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
10. DiBartola SP, Rutgers HC, Zack PM, Tarr MJ. Clinicopathologic findings associated with chronic renal disease in cats: 74 cases (1973–1984). *J Am Vet Med Assoc* 1987; 190: 1196–1202. [[PubMed](#)] [[Google Scholar](#)]
11. Gabor LJ, Canfield PJ, Malik R. Haematological and biochemical findings in cats in Australia with lymphosarcoma. *Aust Vet J* 2000; 78: 456–461. [[DOI](#)] [[PubMed](#)] [[Google](#)]

12. van Hoek I, Daminet S. Interactions between thyroid and kidney function in pathological conditions of these organ systems: a review. *Gen Comp Endocrinol* 2009; 160: 205–215. [\[DOI\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#) 
13. van Hoek I, Daminet S, Notebaert S, Janssens I, Meyer E. Immunoassay of urinary retinol binding protein as a putative renal marker in cats. *J Immunol Methods* 2008; 329: 208–213. [\[DOI\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#) 
14. van Hoek I, Lefebvre HP, Peremans K, Meyer E, Croubels S, Vandermeulen E, et al. Short- and long-term follow-up of glomerular and tubular renal markers of kidney function in hyperthyroid cats after treatment with radioiodine. *Domest Anim Endocrinol* 2009; 36: 45–56. [\[DOI\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#) 
15. Williams TL, Peak KJ, Brodbelt D, Elliott J, Syme HM. Survival and the development of azotemia after treatment of hyperthyroid cats. *J Vet Intern Med* 2010; 24: 863–869. [\[DOI\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#) 
16. Poli A, Tozon N, Guidi G, Pistello M. Renal alterations in feline immunodeficiency virus (FIV)-infected cats: a natural model of lentivirus-induced renal disease changes. *Viruses* 2012; 4: 1372–1389. [\[DOI\]](#) [\[PMC free article\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#) 
17. Glick AD, Horn RG, Holscher M. Characterization of feline glomerulonephritis associated with viral-induced hematopoietic neoplasms. *Am J Pathol* 1978; 92: 321–332. [\[PMC free article\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#) 
18. Ishida T, Washizu T, Toriyabe K, Motoyoshi S, Tomoda I, Pedersen NC. Feline immunodeficiency virus infection in cats in Japan. *J Am Vet Med Assoc* 1989; 194: 221–225. [\[PubMed\]](#) [\[Google Scholar\]](#) 
19. Baxter KJ, Levy JK, Edinboro CH, Vaden SL, Tompkins MB. Renal disease in cats infected with feline immunodeficiency virus. *J Vet Intern Med* 2012; 26: 238–243. [\[DOI\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#) 
20. Woo PC, Lau SK, Wong BH, Fan RY, Wong AY, Zhang AJ, et al. Feline morbillivirus, a previously undescribed paramyxovirus associated with tubulointerstitial nephritis in domestic cats. *Proc Natl Acad Sci USA* 2012; 109: 5435–5440. [\[DOI\]](#) [\[PMC free article\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#) 

21. Lappin MR, Basaraba RJ, Jensen WA. Interstitial nephritis in cats inoculated with Crandell Rees feline kidney cell lysates. *J Feline Med Surg* 2006; 8: 353–356. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
22. Whittemore JC, Hawley JR, Jensen WA, Lappin MR. Antibodies against Crandell Rees feline kidney (CRFK) cell line antigens, alpha-enolase, and annexin A2 in vaccinated and CRFK hyperinoculated cats. *J Vet Intern Med* 2010; 24: 306–313. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
23. DiBartola SP, Buffington CA, Chew DJ, McLoughlin MA, Sparks RA. Development of chronic renal disease in cats fed a commercial diet. *J Am Vet Med Assoc* 1993; 202: 744–751. [[PubMed](#)] [[Google Scholar](#)]
24. Hughes KL, Slater MR, Geller S, Burkholder WJ, Fitzgerald C. Diet and lifestyle variables as risk factors for chronic renal failure in pet cats. *Prev Vet Med* 2002; 55: 1–15. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
25. Reynolds BS, Chetboul V, Nguyen P, Testault I, Concordet D, Carlos Sampedrano C, et al. Effects of dietary salt intake on renal function: a 2 year study in healthy aged cats. *J Vet Intern Med* 2013; 27: 507–515. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
26. Elliott J, Barber PJ. Feline chronic renal failure: clinical findings in 80 cases diagnosed between 1992 and 1995. *J Small Anim Pract* 1998; 39: 78–85. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
27. Polzin DJ. Chronic kidney disease. In: Bartges J, Polzin DJ. (eds). *Nephrology and urology of small animals*. Ames: Wiley-Blackwell, 2011, pp 433–471. [[Google Scholar](#)]
28. Lawler DF, Evans RH, Chase K, Ellersieck M, Li Q, Larson BT, et al. The aging feline kidney: a model mortality antagonist? *J Feline Med Surg* 2006; 8: 363–371. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
29. Weiss DJ, Gagne JM, Armstrong PJ. Relationship between inflammatory hepatic disease and inflammatory bowel disease, pancreatitis, and nephritis in cats. *J Am Vet Med Assoc* 1996; 209: 1114–1116. [[PubMed](#)] [[Google Scholar](#)]
30. Jepson RE, Brodbelt D, Vallance C, Syme HM, Elliott J. Evaluation of predictors of the development of azotemia in cats. *J Vet Intern Med* 2009; 23: 806–813. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

31. Syme HM, Markwell PJ, Pfeiffer D, Elliott J. Survival of cats with naturally occurring chronic renal failure is related to severity of proteinuria. *J Vet Intern Med* 2006; 20: 528–535. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
32. Buffington CA. Invited commentary on ‘The aging feline kidney: a model mortality antagonist?’ *J Feline Med Surg* 2006; 8: 361–362. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
33. Boyd LM, Langston C, Thompson K, Zivin K, Imanishi M. Survival in cats with naturally occurring chronic kidney disease (2000–2002). *J Vet Intern Med* 2008; 22: 1111–1117. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
34. White JD, Norris JM, Baral RM, Malik R. Naturally-occurring chronic renal disease in Australian cats: a prospective study of 184 cases. *Aust Vet J* 2006; 84: 188–194. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
35. Jepson RE. Feline systemic hypertension: classification and pathogenesis. *J Feline Med Surg* 2011; 13: 25–34. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
36. Chetboul V, Lefebvre HP, Pinhas C, Clerc B, Boussouf M, Pouchelon JL. Spontaneous feline hypertension: clinical and echocardiographic abnormalities, and survival rate. *J Vet Intern Med* 2003; 17: 89–95. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
37. Jepson RE, Elliott J, Brodbelt D, Syme HM. Effect of control of systolic blood pressure on survival in cats with systemic hypertension. *J Vet Intern Med* 2007; 21: 402–409. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
38. Chakrabarti S, Syme HM, Brown CA, Elliott J. Histomorphometry of feline chronic kidney disease and correlation with markers of renal dysfunction. *Vet Pathol* 2013; 50: 147–155. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
39. Taugner E, Baatz G, Nobiling R. The renin-angiotensin system in cats with chronic renal failure. *J Comp Pathol* 1996; 115: 239–252. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
40. Elliott J, Rawlings JM, Markwell PJ, Barber PJ. Survival of cats with naturally occurring chronic renal failure: effect of dietary management. *J Small Anim Pract* 2000; 41: 235–242. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
41. Gouni V, Chetboul V, Pouchelon JL, Sampedrano CC, Maurey C, Lefebvre HP. Azotemia in cats with feline hypertrophic cardiomyopathy: prevalence and relationships with

echocardiographic variables. J Vet Cardiol 2008; 10: 117–123. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

42. Javadi S, Djajadiningrat-Laanen SC, Kooistra HS, van Dongen AM, Voorhout G, van Sluijs FJ, et al. Primary hyperaldosteronism, a mediator of progressive renal disease in cats. Domest Anim Endocrinol 2005; 28: 85–104. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

43. Andrew R, Harvey AM, Tasker S. Primary hyperaldosteronism in the cat: a series of 13 cases. J Feline Med Surg 2005; 7: 173–182. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

44. Kelly DF, Lucke VM, McCullagh KG. Experimental pyelonephritis in the cat. 1 Gross and histologic findings. J Comp Pathol 1979; 89: 125–139. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

45. Mayer-Roenne B, Goldstein RE, Erb HN. Urinary tract infections in cats with hyperthyroidism, diabetes mellitus and chronic kidney disease. J Feline Med Surg 2007; 9: 124–132. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

46. King JN, Gunn-Moore DA, Tasker S, Gleadhill A, Strehlau G; Benazepril in Renal insufficiency in Cats Study Group. Tolerability and efficacy of benazepril in cats with chronic kidney disease. J Vet Intern Med 2006; 20: 1054–1064. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

47. Bartges JW, Barsanti JA. Bacterial urinary tract infections in cats. In: Kirk RW. (ed). Current veterinary therapy. 13th ed. Philadelphia: WB Saunders, 2000, pp 880–882. [[Google Scholar](#)]

48. Ichii O, Yabuki A, Sasaki N, Otsuka S, Ohta H, Yamasaki M, et al. Pathological correlations between podocyte injuries and renal functions in canine and feline chronic kidney diseases. Histol Histopathol 2011; 26: 1243–1255. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

49. Vaden SL. Glomerular disease. Top Companion Anim Med 2011; 26: 128–134. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

50. Sato R, Soeta S, Syuto B, Yamagishi N, Sato J, Naito Y. Urinary excretion of N-acetyl-beta-D-glucosaminidase and its isoenzymes in cats with urinary disease. J Vet Med Sci 2002; 64: 367–371. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

51. Ross LA, Finco DR. Relationship of selected clinical renal function tests to glomerular filtration rate and renal blood flow in cats. Am J Vet Res 1981; 42: 1704–1710. [[PubMed](#)] [[Google Scholar](#)]

52. Elliott J, Syme HM, Reubens E, Markwell PJ. Assessment of acid-base status of cats with naturally occurring chronic renal failure. *J Small Anim Pract* 2003; 44: 65–70. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
53. King JN, Tasker S, Gunn-Moore DA, Strehlau G; BENRiC (Benazepril in Renal insufficiency in Cats) Study Group. Prognostic factors in cats with chronic kidney disease. *J Vet Intern Med* 2007; 21: 906–916. [[PubMed](#)] [[Google Scholar](#)]
54. Liu Y. Cellular and molecular mechanisms of renal fibrosis. *Nat Rev Nephrol* 2011; 7: 684–696. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
55. Arata S, Ohmi A, Mizukoshi F, Baba K, Ohno K, Setoguchi A, Tsujimoto H. Urinary transforming growth factor- β 1 in feline chronic renal failure. *J Vet Med Sci* 2005; 67: 1253–1255. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
56. Sawashima K, Mizuno S, Mizuno-Horikawa Y, Shimada A, Kudo T, Kurosawa T. Expression of α -smooth muscle actin and fibronectin in tubulointerstitial lesions of cats with chronic renal failure. *Am J Vet Res* 2000; 61: 1080–1086. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
57. Yabuki A, Mitani S, Fujiki M, Misumi K, Endo Y, Miyoshi N, et al. Comparative study of chronic kidney disease in dogs and cats: induction of myofibroblasts. *Res Vet Sci* 2010; 88: 294–299. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
58. Brown SA. Oxidative stress and chronic kidney disease. *Vet Clin North Am Small Anim Pract* 2008; 38: 157–166. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
59. Keegan RF, Webb CB. Oxidative stress and neutrophil function in cats with chronic renal failure. *J Vet Intern Med* 2010; 24: 514–519. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
60. Jensen J, Henik RA, Brownfield M, Armstrong J. Plasma renin activity and angiotensin I and aldosterone concentrations in cats with hypertension associated with chronic renal disease. *Am J Vet Res* 1997; 58: 535–540. [[PubMed](#)] [[Google Scholar](#)]
61. Mishina M, Watanabe T, Fujii K, Maeda H, Wakao Y, Takahashi M. Non-invasive blood pressure measurements in cats: clinical significance of hypertension associated with chronic renal failure. *J Vet Med Sci* 1998; 60: 805–808. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
62. Pedersen KM, Pedersen HD, Häggström J, Koch J, Ersbøll AK. Increased mean arterial pressure and aldosterone-to-renin ratio in Persian cats with polycystic kidney disease. *J Vet Intern Med* 2003; 17: 21–27. [[PubMed](#)] [[Google Scholar](#)]

63. Watanabe T, Mishina M. Effects of benazepril hydrochloride in cats with experimentally induced or spontaneously occurring chronic renal failure. *J Vet Med Sci* 2007; 69: 1015–1023. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
64. Lefebvre HP, Toutain PL. Angiotensin-converting enzyme inhibitors in the therapy of renal diseases. *J Vet Pharmacol Ther* 2004; 27: 265–281. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
65. Brown SA, Brown CA, Jacobs G, Stiles J, Hendi RS, Wilson S. Effects of the angiotensin converting enzyme inhibitor benazepril in cats with induced renal insufficiency. *Am J Vet Res* 2001; 62: 375–383. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
66. Schrier RW, Masoumi A, Elhassan E. Aldosterone: role in edematous disorders, hypertension, chronic renal failure, and metabolic syndrome. *Clin J Am Soc Nephrol* 2010; 5: 1132–1140. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
67. Brown SA, Brown CA. Single-nephron adaptations to partial renal ablation in cats. *Am J Physiol* 1995; 269: R1002–R1008. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
68. Brown SA, Crowell WA, Brown CA, Barsanti JA, Finco DR. Pathophysiology and management of progressive renal disease. *Vet J* 1997; 154: 93–109. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
69. Ross LA, Finco DR, Crowell WA. Effect of dietary phosphorus restriction on the kidneys of cats with reduced renal mass. *Am J Vet Res* 1982; 43: 1023–1026. [[PubMed](#)] [[Google Scholar](#)]
70. Adams LG, Polzin DJ, Osborne CA, O'Brien TD, Hostetter TH. Influence of dietary protein/calorie intake on renal morphology and function in cats with 5/6 nephrectomy. *Lab Invest* 1994; 70: 347–357. [[PubMed](#)] [[Google Scholar](#)]
71. Finco DR, Brown SA, Brown CA, Crowell WA, Sunvold G, Cooper TL. Protein and calorie effects on progression of induced chronic renal failure in cats. *Am J Vet Res* 1998; 59: 575–582. [[PubMed](#)] [[Google Scholar](#)]
72. Elliott J, Syme HM, Markwell PJ. Acid-base balance of cats with chronic renal failure: effect of deterioration in renal function. *J Small Anim Pract* 2003; 44: 261–268. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
73. Chakrabarti S, Syme HM, Elliott J. Clinicopathological variables predicting progression of azotemia in cats with chronic kidney disease. *J Vet Intern Med* 2012; 26: 275–281.

[\[DOI\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)

74. Kuwahara Y, Ohba Y, Kitoh K, Kuwahara N, Kitagawa H. Association of laboratory data and death within one month in cats with chronic renal failure. *J Small Anim Pract* 2006; 47: 446–450. [\[DOI\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)

75. Brown S, Atkins C, Bagley R, Carr A, Cowgill L, Davidson M, et al. Guidelines for the identification, evaluation, and management of systemic hypertension in dogs and cats. *J Vet Intern Med* 2007; 21: 542–558. [\[DOI\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)

76. Kidder A, Chew D. Treatment options for hyperphosphataemia in feline CKD. *J Feline Med Surg* 2009; 11: 913–924. [\[DOI\]](#) [\[PMC free article\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)

77. Barber PJ, Elliott J. Feline chronic renal failure: calcium homeostasis in 80 cases diagnosed between 1992 and 1995. *J Small Anim Pract* 1998; 39: 108–116. [\[DOI\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)

78. Finch NC, Geddes RF, Syme HM, Elliott J. Fibroblast growth factor 23 (FGF-23) concentrations in cats with early nonazotemic chronic kidney disease (CKD) and in healthy geriatric cats. *J Vet Intern Med* 2013; 27: 227–233. [\[DOI\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)

79. Geddes RF, Finch NC, Elliott J, Syme HM. Fibroblast growth factor 23 in feline chronic kidney disease. *J Vet Intern Med* 2013; 27: 234–241. [\[DOI\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)

80. Bertazzolo W, Toscani L, Calcaterra S, Crippa L, Caniatti M, Bonfanti U. Clinicopathological findings in five cats with paw calcification. *J Feline Med Surg* 2003; 5: 11–17. [\[DOI\]](#) [\[PMC free article\]](#) [\[PubMed\]](#) [\[Google Scholar\]](#)

81. DiBartola SP, Autran De Morais H. Disorders of potassium: hypokalemia and hyperkalemia. In: DiBartola SP. (ed). *Fluid, electrolyte, and acid-base disorders in small animal practice*. 4th ed. St Louis: Elsevier Saunders, 2012, pp 92–119. [\[Google Scholar\]](#)

82. Dow SW, Fettman MJ, Curtis CR, LeCouteur RA. Hypokalemia in cats: 186 cases (1984–1987). *J Am Vet Med Assoc* 1989; 194: 1604–1608. [\[PubMed\]](#) [\[Google Scholar\]](#)

83. Adams LG, Polzin DJ, Osborne CA, O'Brien TD. Effects of dietary protein and calorie restriction in clinically normal cats and in cats with surgically induced chronic renal failure. *Am J Vet Res* 1993; 54: 1653–1662. [\[PubMed\]](#) [\[Google Scholar\]](#)

84. Dow SW, Fettman MJ, Smith KR, Hamar DW, Nagode LA, Refsal KR, et al. Effects of dietary acidification and potassium depletion on acid-base balance, mineral metabolism

and renal function in adult cats. J Nutr 1990; 120: 569–578. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

85. Cowgill LD. Pathophysiology and management of anemia in chronic progressive renal failure. Semin Vet Med Surg (Small Anim) 1992; 7: 175–182. [[PubMed](#)] [[Google Scholar](#)]

86. Chalhoub S, Langston C, Eatroff A. Anemia of renal disease: what it is, what to do and what's new. J Feline Med Surg 2011; 13: 629–640. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

87. Lees GE, Brown SA, Elliott J, Grauer GE, Vaden SL. Assessment and management of proteinuria in dogs and cats: 2004 ACVIM Forum Consensus Statement (small animal). J Vet Intern Med 2005; 19: 377–385. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

88. Syme HM. Proteinuria in cats. Prognostic marker or mediator? J Feline Med Surg 2009; 11: 211–218. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

89. Bartlett PC, Van Buren JW, Bartlett AD, Zhou C. Case-control study of risk factors associated with feline and canine chronic kidney disease. Vet Med Int 2010; Article ID: 957570. DOI: 10.4061/2010/957570. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]

90. Goldstein RE, Marks SL, Kass PH, Cowgill LD. Gastrin concentrations in plasma of cats with chronic renal failure. J Am Vet Med Assoc 1998; 213: 826–828. [[PubMed](#)] [[Google Scholar](#)]

91. Schulman RL, Krawiec DR. Gastrointestinal complications of uremia. In: Bonagura JD. (ed). Kirk's current veterinary therapy. XIII Small animal practice. Philadelphia: WB Saunders, 2000, pp 864–866. [[Google Scholar](#)]